# MILITARY SPECIFICATION

## BOMB, FIRE, 750-POUND, M116A2

## (W/O FUZE COMPONENTS AND FILLING)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers an incendiary bomb.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-T-5021 - Test; Aircraft Welding Operator's Certification.
MIL-I-6868 - Inspection Process, Magnetic Particle.

STANDARDS

MILITARY

MIL-STD-20 - Welding Terms and Definitions.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.



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DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

D14-5-2702 - Bomb, Fire, 750-lb., M116A2 - Metal Parts - Bill of Material (2 sheets).
D14-17-907 - Packing, 750-lb. Fire bomb, M116A2 - Bill of Material.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing D14-5-2702 or on the subsidiary drawings shall conform to the specifications thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the bomb shall conform to the specifications and drawings listed on Drawing D14-5-2702 and subsidiary drawings.

3.2 Assembly. The bomb shall be assembled as shown on Drawing E14-5-2703 by connecting the nose, center and tail sections together using a torque of 130 + 5 inch-pounds on the locking bolts.

3.2.1 Dents. The maximum permissible depth of any dent on the skin of the bomb section shall not exceed 1/16-inch per 1/4-inch of width or diameter.

3.3 Leakage. The bomb shall not leak when tested in accordance with 4.4.4.1.

3.4 Slosh and vibration. The bomb shall not leak nor show any evidence of structural failure when tested in accordance with 4.4.4.2.

3.5 Suspension lug. The suspension lug shall be magnetically inspected for defects in accordance with MIL-I-6868.



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3.6 Fusion welding.

3.6.1 Recorded welding procedure. The supplier shall establish and qualify a welding procedure in accordance with 3.6.1.1 covering each welding operation to be performed, and maintain the procedure throughout production. Approval of a welding procedure makes no warranty that bombs fabricated in accordance with the approved procedure will comply with the other requirements of this specification. The procedure shall include, but not be limited to, the following information as applicable:

(a) Sketch of joint indicating edge preparation and weld size, fit and gage of parts being joined, required penetration, etc.
(b) Manufacturer, type and serial number of welding machine.
(c) Type of torch.
(d) Gas cup diameter.
(e) Description of electrode or filler rod.
(f) Inert gas composition and flow rate.
(g) Current amperes, alternating current (ac) or direct current (dc).
(h) Archvoltage (where applicable).
(i) Wire or electrode feed rate.
(j) Preheat (yes or no).
(k) Number of passes.
(l) Anti-splatter compound.
(m) Backing.
(n) Cleaning process.
(o) Welding speed.
(p) Welding quality control procedure.
(q) Other specific data (repair procedures, etc.).

3.6.1.1 Procedure qualification. Prior to qualification of procedures, welders, or welding operators, the supplier shall submit a proposed welding procedure to the contracting officer for review and tentative approval. The supplier shall demonstrate to the selected representatives of the contracting officer the satisfactory quality of the welds produced using this proposed procedure by fabricating a minimum of five samples of each different type joint which meet the strength, fusion and workmanship requirements of this specification. The proposed welding procedure shall be used in fabricating the preproduction sample. Final approval of the proposed procedure shall be given by the contracting officer upon preproduction sample acceptance.

3.6.2 Welder or welding operator qualification. All manual fusion welding shall be performed by welders who have met the requirements of MIL-T-5021 for group IV, class A welding using the established welding



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procedure. Welding operators shall be considered qualified to weld with the established welding procedure when they have demonstrated their ability to produce welds meeting the requirements of this specification when using that procedure. The terms "welder" and "welding operator" shall be as specified in MIL-STD-20.

3.6.3 Welding workmanship samples. The supplier shall prepare samples of each welded joint clearly indicating the workmanship quality limits of welds to be maintained during production. These shall be submitted with the proposed welding procedure and be retained by the inspector.

3.6.4 Production welding. All production welding shall be performed in accordance with the established welding procedure and be controlled to meet approved workmanship standards. Any change to the established procedure shall be qualified in accordance with the applicable portion of 3.6.1.1 prior to its incorporation. If, for any reason, the contracting officer's representative doubts the capability of any welding apparatus to function satisfactorily, he shall require welding operators certification tests as specified in MIL-T-5021, applicable to the type of work for which the equipment is intended. If, under these conditions, the applicable certification requirements cannot be met, the equipment shall not be used until the necessary repairs, adjustments, or replacements have been made. The ultimate tensile strength of the sample welded joints shall be equal to or greater than that of the adjoining parent metal.

3.7 Preproduction. Prior to the start of regular production a preproduction sample of bombs shall be produced in accordance with this specification for examination and tests.

3.8 Workmanship. The bomb shall be free from foreign matter and damage such as dents, (other than those in 3.2.1) visual cracks or deformation which would impair its use.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.



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4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 25 bombs shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. The sample subassemblies of the bomb shall be examined in accordance with the classification of defects. The AQL's do not apply.

4.3.2.2 For nondestructive tests. Ten assembled bombs shall be randomly taken from the preproduction sample and tested in accordance with 4.4.4.1, and three bombs shall be tested in accordance with 4.4.4.2.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The acceptance number is 0. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the bombs of one interfix number, produced by one manufacturer, at one plant, in not more than six consecutive work shifts, during one calendar week, from the same materials and under essentially the same manufacturing conditions.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For nondestructive test . One representative bomb shall be randomly taken from a block of 30 consecutively produced bombs.



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4.4.3 Inspection procedure.

4.4.3.1 For examination. The sample subassemblies of the bomb shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.4.3.2 For nondestructive test. The sample bomb shall be tested in accordance with 4.4.4.1. The acceptance number is 0.

4.4.3.3 Classification of defects.

(a) Center-assembly (Dwg E14-5-2704).

| Categories | Defects |
|---|---|
| Critical: | None defined |
| Major: | AQL 1.0 percent defective |
| 101 | Component incorrect or incorrectly assembled |
| 102 | Dents (3.2.1) |
| 103 | Overall length incorrect |
| 104 | Suspension lug opening height incorrect |
| 105 | Alignment marking missing or incorrectly located |
| 106 | Distance between suspension lugs incorrect |
| 107 | Suspension lug torque incorrect |
| 108 | Welding incorrect (3.6.4) |
| 109 | Suspension lug defects (3.5) |
| 110 | Marking missing, illegible or incorrect |
| 111 | Workmanship (3.8) |

(b) Nose assembly (Dwg E14-5-2101).

| Categories | Defects |
|---|---|
| Critical: | None defined |
| Major: | AQL 1.0 percent defective |
| 101 | Component incorrect or incorrectly assembled |
| 102 | Nose cap guide nonfunctioning |
| 103 | Rolled threads incorrect |
| 104 | Nose cap plunger nonfunctioning |
| 105 | Nose tube clogged |
| 106 | Welding incorrect (3.6.4) |
| 107 | Marking missing, illegible or incorrect |
| 108 | Overall length incorrect |
| 109 | Workmanship (3.8) |



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4.4.3.3 Classification of defects (continued).

(c) Tail-assembly (Dwg E14-5-2103).

| Categories | | Defects |
|---|---|---|
| Critical: | | None defined |
| Major: | AQL 1.0 percent defective | |
| 101 | | Component incorrect or incorrectly assembled |
| 102 | | Rolled thread incorrect |
| 103 | | Tail tube clogged |
| 104 | | Welding incorrect (3.6.4) |
| 105 | | Marking missing, illegible or incorrect |
| 106 | | Overall length incorrect |
| 107 | | Workmanship (3.8) |

(d) Plunger, tail cone-assembly or plunger tail cone (Dwg C14-5-1568 or C14-5-2174).

| Categories | | Defects |
|---|---|---|
| Critical: | | None defined |
| Major: | AQL 1.0 percent defective | |
| 101 | | Component incorrect or incorrectly assembled |
| 102 | | Thread incorrect |
| 103 | | Hole diameter incorrect |
| 104 | | Plating missing or inadequate |
| 105 | | Workmanship (3.8) |

(e) Cap, nose-assembly (Dwg D14-5-2170).

| Categories | | Defects |
|---|---|---|
| Critical: | | None defined |
| Major: | AQL 1.0 percent defective | |
| 101 | | Component incorrect or incorrectly assembled |
| 102 | | Marking missing, illegible or incorrect |
| 103 | | Workmanship (3.8) |



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4.4.3.3 Classification of defects (continued).

(f) Cone, tail-assembly (Dwg D14-5-2172).

| Categories | | Defects |
|---|---|---|
| Critical: | | None defined |
| Major: | AQL 1.0 percent defective | |
| 101 | | Component incorrect or incorrectly assembled |
| 102 | | Marking missing, illegible or incorrect |
| 103 | | Workmanship (3.8) |

(g) Preparation for delivery (Section 5).

| Categories | | Defects |
|---|---|---|
| Critical: | | None defined |
| Major: | AQL 2.5 percent defective | |
| 101 | | Preservative missing |
| 102 | | Containers incorrect |
| 103 | | Bomb component, subassembly or instruction card missing or incorrectly located in container |
| 104 | | Filler pads missing or insufficient for tight pack |
| 105 | | Closure incorrect |
| 106 | | Containers damaged |
| 107 | | Strapping loose or missing |
| 108 | | Marking missing, illegible or incorrect |

4.4.4 Tests.

4.4.4.1 Leakage. The bomb surface shall be brushed or sprayed with a soap solution and subjected to an internal air pressure of 5 + 1/2 pound per square inch gage (psig). As an alternative, the assembly shall be submerged to a depth of 4 to 6 inches, from uppermost surface for a minimum period of 1 minute in water containing a suitable wetting agent of sufficient concentration to prevent air bubbles from clinging to the surfaces. The submerged bomb shall be observed for air bubbles other than trapped air bubbles. The sealing washers under the locking bolts shall be replaced following each test. If leakage is attributed to a defective sealing washer under a locking bolt or to incorrect assembly by test personnel which results in a damaged gasket, the washer or gasket shall be replaced and the bomb retested.



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4.4.4.2 Slosh and vibration. The bomb shall be filled as specified in 4.4.4.3 and then suspended and steadied by sway braces from an "A" frame or other suitable fixture (see fig. 1) simulating the bomb installation on an airplane. Each of the four sway brace pads shall be circular with a flat contact face of 2 + 1/4 square inch area. The sway brace pads shall be positioned 16 inches on center (one inch fore and aft of the lugs) and equally tested with respect to the lugs radially at 22-1/2 + 1/2 degree or a total included angle of 45 + 1 degree. The "A" frame or fixture shall be bolted to the platform of a suitable slosh and vibration apparatus and subjected to the tests specified in 4.4.4.3 and 4.4.4.4. In the event of a mechanical failure of the slosh and vibration equipment, the bomb tested shall be drained and inspected for evidence of structural failure. Upon repair of mechanical defects in the equipment, the bomb shall be again placed on the test table (providing it has not failed the test) and the testing continued to its completion.

4.4.4.3 Combined slosh and vibration. The bomb shall be filled with 75 + 5 gallons of water containing a suitable dye to indicate leakage. The bomb shall then be subjected to the following conditions of combined slosh and vibration for a period of 10 + 1/4 hour.

(a) Slosh. The bomb shall be rocked in the plane of its longitudinal axis 16 to 20 cycles a minute through a total angle of 30 degrees, approximately 15 degrees to each side of the horizontal position.

(b) Vibration. The eccentrics of the test table shall be adjusted such that an average peak to peak vibration displacement of 0.020 + 0.0005 inch result adjacent to the suspension lugs. Measurements of peak to peak vibration displacement shall be taken at two points on the top adjacent to the suspension lugs and two points on the bottom adjacent to the bulkheads. The frequency of vibrating shall be 2,000 - 60 cycles a minute.

4.4.4.4 Vibration. The bomb tested as specified in 4.4.4.3, shall be completely filled with water containing dye and subjected to vibration alone, for a period of 10 + 1 minute, with the eccentrics maintained at the settings previously established for the slosh and vibration test (see 4.4.4.3(b)).

## 5. PREPARATION FOR DELIVERY

5.1 Cleaning and drying. Unpainted metal surfaces including threaded areas shall be cleaned using method C-3 of MIL-P-116 and any of the drying methods listed therein.



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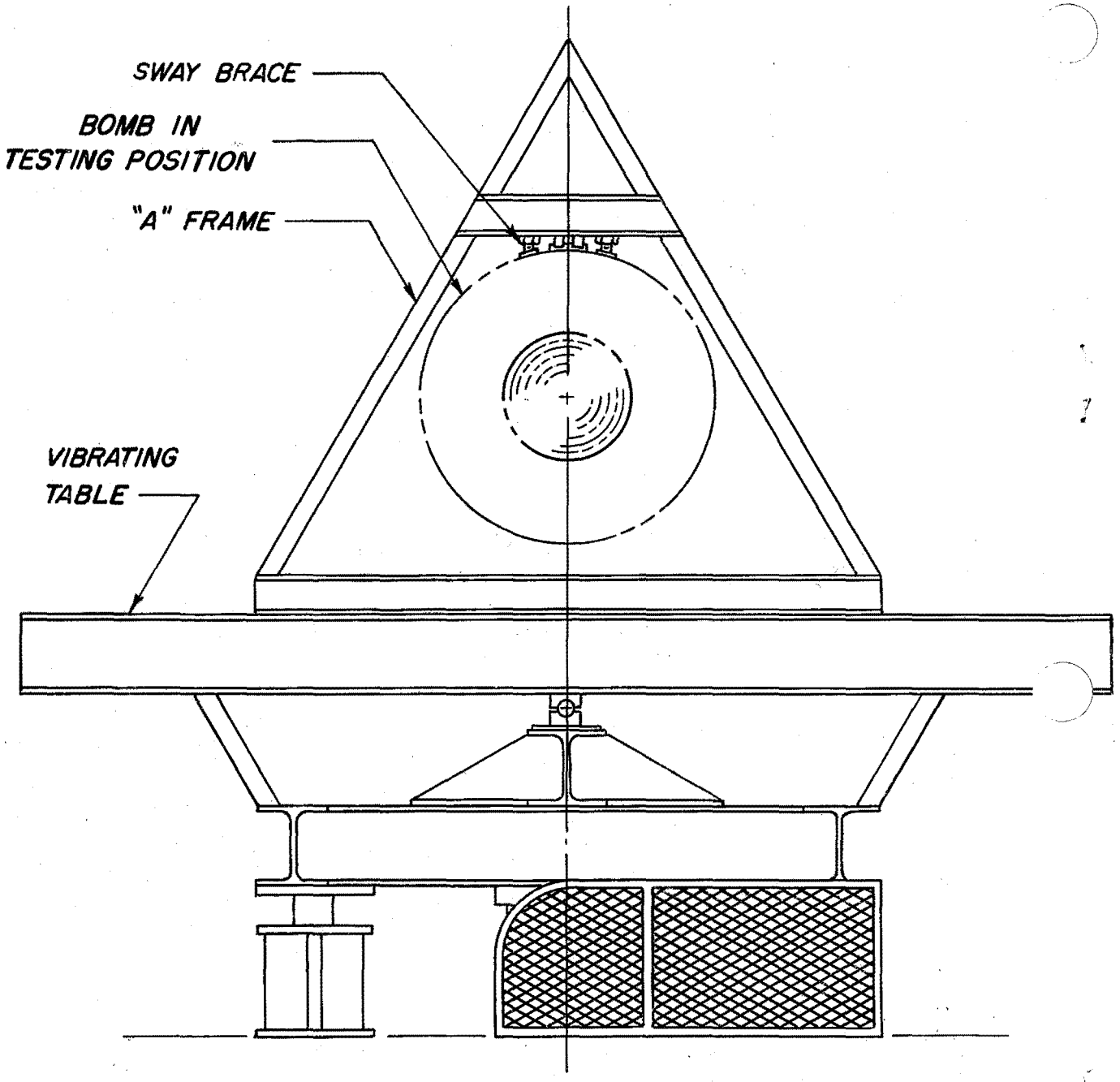



FIGURE 1

SLOSH AND VIBRATION TESTING APPARATUS.



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5.2 Preservation and packaging.

5.2.1 Level A. The bomb cap, cone, instruction card and accessories shall be preserved and packaged in accordance with Drawing C14-17-662.

5.3 Packing, level A. The bomb components shall be packed for shipment in accordance with Drawing D14-17-908.

5.4 Marking. In addition to any special marking in conjunction with information furnished in the contract or order, marking shall be in accordance with the applicable requirements of Drawing D150-2-1.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

Custodians:

Army - MU
Air Force - 70

User activity:

Army - SM

Preparing activity:

Army - MU(EA)

Project No. 1325-0084



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